

MINI NLP PROJECT REPORT

NLP-Based Spam Message Detection Using TF-IDF and Machine Learning

A Mini Project Report

Submitted by

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Project developed using Python, NLTK, TF-IDF and Machine Learning

1. Problem Statement

Spam messages are unwanted messages that may contain advertisements, fraudulent offers, misleading information, or malicious links. With the increasing use of SMS communication, automatically identifying spam messages has become important. This project develops an NLP-based machine learning system that classifies SMS messages as Spam or Ham using text preprocessing, TF-IDF feature extraction, and machine learning classification algorithms.

2. Objectives

- To develop an NLP-based system for detecting spam SMS messages.
- To preprocess textual SMS data using NLP techniques.
- To convert text messages into numerical features using TF-IDF.
- To train machine learning models for spam classification.
- To compare the performance of Multinomial Naive Bayes and Logistic Regression.
- To evaluate the models using accuracy, precision, recall, and F1-score.
- To develop a system capable of predicting whether a new message is spam or ham.

3. Dataset

The project uses an SMS spam classification dataset containing 5,572 messages. The dataset consists of two main attributes: label and message. The label identifies whether a message is ham or spam, while message contains the actual SMS text.

Dataset Information	Value
Total messages	5,572
Ham messages	4,825
Spam messages	747
Input attribute	SMS message
Target attribute	Label
Classes	Ham, Spam

The dataset is imbalanced because the number of ham messages is considerably higher than the number of spam messages. This imbalance is considered when interpreting model performance.

4. Methodology

The overall workflow of the project is:

SMS Dataset → Text Preprocessing → TF-IDF → Train/Test Split → Machine Learning → Evaluation → Prediction

4.1 Text Preprocessing

The SMS messages were processed using the following NLP preprocessing steps:

- Conversion of text to lowercase.
- Removal of URLs.
- Removal of numbers.
- Removal of punctuation.
- Tokenization of text.
- Removal of English stopwords.
- Removal of very short words.

4.2 TF-IDF Feature Extraction

TF-IDF (Term Frequency-Inverse Document Frequency) was used to convert the processed text into numerical feature vectors. The implementation used a maximum of 5,000 features and an n-gram range of 1–2. Therefore, both individual words and two-word combinations were considered.

4.3 Machine Learning Models

Two classification algorithms were implemented and compared.

Multinomial Naive Bayes: Naive Bayes was selected because it is efficient and commonly suitable for text classification tasks.

Logistic Regression: Logistic Regression was used as a second classification model to compare its performance with Naive Bayes.

5. Implementation

The project was implemented in Python using Google Colab. The main libraries used were Pandas, NumPy, NLTK, Scikit-learn, Matplotlib, and Seaborn.

The dataset was divided into training and testing sets using an 80:20 split with stratification to preserve the class distribution. TF-IDF was fitted on the training data and then used to transform both training and testing data. Both Multinomial Naive Bayes and Logistic Regression were trained using the resulting TF-IDF features.

6. Results

The models were evaluated using accuracy, precision, recall, and F1-score. The obtained results are shown below.

Model	Accuracy	Precision	Recall	F1 Score
Multinomial Naive Bayes	97.04%	100.00%	77.85%	87.55%
Logistic Regression	96.95%	100.00%	77.18%	87.12%

Multinomial Naive Bayes achieved the highest overall performance among the two models. It achieved 97.04% accuracy, 100% precision, 77.85% recall, and 87.55% F1-score. Since it achieved slightly better results across the reported metrics, Multinomial Naive Bayes was selected as the final model for the spam detection system.

The 100% precision indicates that the messages predicted as spam by the model were correctly classified as spam in the test set. The recall of 77.85% indicates that the model detected a substantial portion of the actual spam messages, although some spam messages were still classified as ham.

7. Limitations

- The dataset contains fewer spam messages than ham messages.
- The dataset may not represent all modern spam-message patterns.
- The system primarily works with English SMS messages.
- TF-IDF represents text numerically but does not fully understand contextual meaning.
- New spam patterns may require additional training data and model updates.
- The system was developed and tested as a prototype rather than as a production-level mobile application.

8. Future Scope

- Use a larger and more recent spam-message dataset.
- Support multiple languages.
- Implement advanced NLP models such as BERT.
- Develop a web or mobile application for real-time spam detection.
- Continuously update the model with new spam examples.
- Add URL and sender-based analysis.
- Improve detection of new and evolving spam patterns.

9. Conclusion

This project demonstrates the application of Natural Language Processing and machine learning for automatic SMS spam detection. Text preprocessing and TF-IDF were used to transform SMS messages into suitable numerical features, after which Multinomial Naive Bayes and Logistic Regression were trained and compared.

Among the two models, Multinomial Naive Bayes performed best with an accuracy of 97.04% and an F1-score of 87.55%. The results demonstrate that traditional NLP and machine learning techniques can provide effective spam-message classification. The project can be further extended using larger datasets, advanced NLP models, multilingual support, and real-time applications.

10. Technologies and Tools Used

- Programming Language: Python
- Development Environment: Google Colab
- NLP Library: NLTK
- Machine Learning: Scikit-learn
- Data Processing: Pandas, NumPy
- Visualization: Matplotlib, Seaborn
- Feature Extraction: TF-IDF
- Classification Models: Multinomial Naive Bayes and Logistic Regression